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An agent-based model for estimating emissions reduction equivalence among leak detection and repair programs

Thomas A. Fox [a, *], Mozhou Gao [a], Thomas E. Barchyn [a], Yorwearth L. Jamin [b], Chris H. Hugenholtz [a]

[a] Centre for Smart Emissions Sensing Technologies, University of Calgary, 2500 University Drive NW, Calgary, AB, T2N 1N4, Canada
[b] Clearstone Engineering Ltd., 900 6 Ave SW #700, Calgary, AB, T2P 3K2, Canada

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ABSTRACT

Alternative leak detection and repair (alt-LDAR) programs are being introduced by regulators to provide flexibility in how oil and gas producers manage fugitive methane emissions. However, emissions reduction equivalence must be established between a proposed program and a regulatory standard. We present LDAR-Sim, an open-source, agent-based numerical model for evaluating LDAR program performance. Novel functionality includes the ability to: (1) set facility-specific LDAR requirements, (2) simultaneously deploy multiple technologies, (3) integrate screening and close-range methods, (4) include environmental constraints, and (5) explore the impact of design emissions on screening technologies. We demonstrate LDAR-Sim in two case studies with diverse LDAR programs deployed on real assets. We find that equivalence determinations depend on explicit definition of reference standards, including weather and labour availability. Screening program performance is sensitive to the confounding presence of design emissions and to decision-making strategies that guide follow-up inspections. When comparing programs, differences in simulated performance are sensitive to leak production and null repair rates, two elusive parameters used in previous studies. Future work should better constrain empirical inputs and validate specific LDAR programs as alt-LDAR deployment increases. Regulators and producers can use LDAR-Sim to facilitate the adoption of emerging technologies but should be aware of its limitations.
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1. Introduction

Oil and gas systems are the largest source of anthropogenic methane emissions in Canada and the US (EPA, 2019). In the upstream O&G sector, methane can be emitted by design (i.e. intentional vented emissions) or as accidental leaks (i.e. unintentional fugitive emissions). To address fugitive emissions, regulators often require producers to implement leak detection and repair (LDAR) programs (Alberta Energy Regulator, 2018; Government of Canada, 2018). Most LDAR programs consist of periodic surveys using handheld instruments such as optical gas imaging (OGI) cameras or organic vapour analyzers. These 'conventional' surveys are conducted at the component scale using close-range methods with high sensitivity.

Alternative approaches to LDAR (alt-LDAR) may exist (Schwietzke et al., 2019; Fox et al., 2019a). We define alt-LDAR as any deviation from a prescribed regulatory LDAR program (reg-LDAR). Over the past decade, a variety of technologies, methods, and program designs have emerged that may contribute to improving the cost- and mitigation-effectiveness of LDAR. In response, various regulatory agencies in the US and Canada have moved to recognize alt-LDAR as a potentially viable path to compliance (Alberta Energy Regulator, 2018; CDPHE, 2018). Regulators often require evidence that proposed alt-LDAR programs achieve equivalent mitigation to the reg-LDAR standard. A combination of controlled testing, simulation modeling, and field trials has been recommended (Fox et al., 2019b).

Simulation modeling provides several advantages as a method of understanding the emissions reductions potential of an LDAR program. First, simulations allow a long-term estimation of emissions dynamics. Second, direct testing of LDAR tools on real leaks is slow, expensive, and possibly unfair as a representative distribution of emission rates is difficult to acquire (Brandt et al., 2016). Third,

* Corresponding author.
E-mail address: thomas.fox@ucalgary.ca (T.A. Fox).

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empirically measuring total mitigation progress is challenging due to the high uncertainties typical of most quantification methods (Ravikumar et al., 2019). The first and only existing peer-reviewed LDAR simulation tool was the Fugitive Emissions Abatement Simulation Testbed (FEAST) (Kemp et al., 2016). FEAST generates a virtual leak field in which leaks arise and are removed stochastically at the component scale following a Markov process. Detection modules are then applied to the leak field. When a survey is conducted at a component, detection is evaluated by simulating a Gaussian plume and considering the physical properties of the sensor. If detected, leaks are repaired and the component's Markov state switches to not emitting. Mitigation is calculated as the difference between time-integrated emissions under the LDAR program and a 'null' scenario that estimates baseline emissions in the absence of LDAR.

The original version of FEAST is a valuable tool but has drawbacks that may limit its broader utility. First, the success of an LDAR program depends not only upon the theoretical detection capabilities of a method, but also upon practical considerations that vary by producer, facility, environment, and policy context. For example, an aircraft-based remote sensing system may perform poorly in high-latitude winters due to snow cover, cloud cover, and low solar irradiance. An alt-LDAR program that commits to monthly surveys using this system may only be successful in specific geographical or seasonal contexts. If only a handful of days are suitable for deployment, labour and equipment availability may become limited, especially if only one aircraft or crew exists. Second, LDAR simulation tools should provide producers with an opportunity to explore and optimize across a range of possible programs. For example, a producer may propose a program that simultaneously integrates aircraft, trucks, and OGIs with facility-specific survey schedules and frequencies, and custom repair protocols. Important interaction effects may emerge from concurrent deployments that could influence the equivalence determination.

To meet these needs, we present LDAR-Sim, an open-source agent-based modeling framework that permits the simultaneous deployment of multiple methods, each with multiple agents (crews), while allowing for the definition of custom LDAR programs. We first present a detailed description of LDAR-Sim, then demonstrate the model with a case study using real assets in Alberta, Canada. We conclude with a comprehensive sensitivity analysis (SA) and discussion of modeling assumptions and shortcomings, and their implications for generating equivalence determinations.

2. Model description

LDAR-Sim is written in Python 3 and uses the *initialize-update-finalize* Basic Model Interface (BMI) of the Community Surface Dynamics Modeling System (Peckham et al., 2013). The BMI provides a standardized, modular, and intuitive model structure to facilitate collaboration, accelerate learning and streamline coupling with external software. In LDAR-Sim, each LDAR program consists of a set of assets, one or more leak detection methods, and a set of rules governing deployment and repair. Assets are combined with empirical inputs to initialize a leak field. A dynamic simulation with daily timesteps and minute-scale temporal resolution is then used to evaluate the program (Fig. 1). Leaks arise stochastically and are detected probabilistically according to method-specific empirical detection data. Aggregate-scale detections (i.e. by screening methods) can produce flags, which trigger follow-up by close-range methods. Leaks detected at close-range are tagged and enter a repair queue. Tagged leaks are removed from the simulation following user-defined reporting delays and repair intervals. Parameterization, selection of empirical inputs, and sensitivity analyses (SA) should be done on a case-by-case basis; no defaults exist.

In this paper, LDAR-Sim is presented in the context of demonstrating emissions reduction equivalence among different LDAR programs. However, LDAR-Sim could be used by various stakeholders for other applications. For example, policymakers could use LDAR-Sim to estimate whether existing or proposed regulations will achieve emissions reduction targets. Regulators might use LDAR-Sim to understand how to deploy different technologies as part of compliance efforts. Technology innovators might also use LDAR-Sim to understand how their solutions compare to existing technologies, to identify ideal environments for deployment, or establish innovation trajectories (e.g., sensor sensitivity requirements).

2.1. Program definition

A primary motivation in developing LDAR-Sim is to enable the precise definition of custom LDAR programs. Defining a program requires identifying the detection method(s), the asset base, and deployment protocols. In practice, each method module should be designed in accordance with the technology and work practice proposed by the solution provider. Ideally, a set of relevant performance metrics are developed through controlled testing. We provide a set of general method templates, including OGI, a mobile ground laboratory (MGL), and an aircraft system. Each method can be parameterized according to relevant criteria, which commonly include detection performance, labour capacity, minimum survey intervals, environmental constraints, reporting delays, and follow-up criteria for screening methods. These templates can also be used to develop modules for other methods.

The quality of the parameterization depends on the empirical data available. For example, detection performance can be a static minimum detection limit or a function of leak size and environmental conditions. Labour capacity is defined by the number of unique agents (crews) that can be deployed, the duration of a workday (including daylight constraints), the time to complete each facility, and time spent between facilities, which combines transit, permitting, breaks, and complications. For reg-LDAR, minimum survey intervals reflect policy requirements designed to spread out surveys. For alt-LDAR, these intervals represent clearly-defined work practices within a proposed program. Reporting delays represent the time interval from detection to actionable information. Delays may be trivial for some methods, but functionality has been included for methods that are slow to process and deliver actionable information.

Environmental conditions can be incorporated in two ways. First, operational envelopes are defined, preventing deployment if conditions are unsuitable (a binary determination as defined by each method's work practice). Second, detection performance within an envelope can be configured as a function of the environment through a custom model relating environment to performance. As environmental conditions vary in space and time, so should the performance of LDAR methods. To account for local environmental context, LDAR-Sim uses gridded ERA-Interim reanalysis data. These data have global daily coverage with ~80 km spatial resolution from the European Centre for Medium-Range Weather Forecasts (Dee et al., 2011). At a minimum, LDAR-Sim requires temperature, wind speed, and precipitation as inputs, as these are known to impact the performance of OGIs (Ravikumar and Brandt, 2017), which are typically used as the reference target in equivalence (Alberta Energy Regulator, 2018; Government of Canada, 2018). However, the reanalysis contains a detailed description of the weather, including wind direction (for downwind plume intersection methods), snow depth and cloud cover

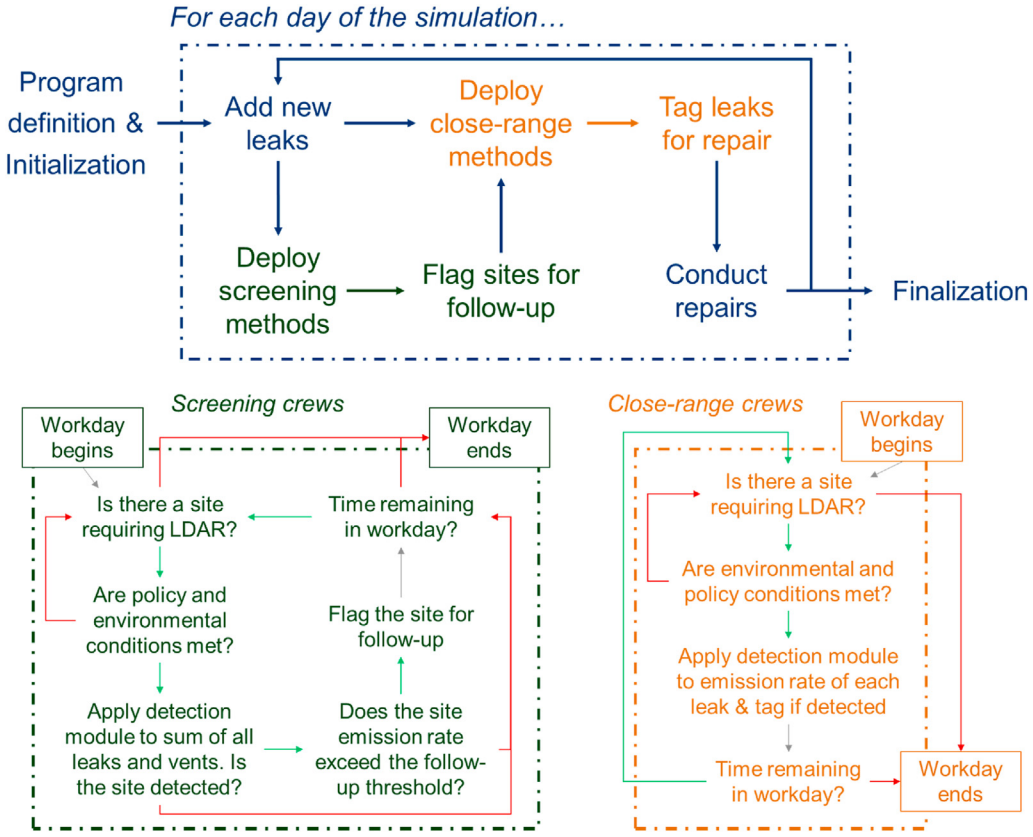


Fig. 1. Overview of LDAR-Sim. Green text is used for screening methods and orange text for close-range methods. Red arrows represent 'no', green arrows are 'yes', and grey arrows are mandatory. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

(for remote sensing methods), and solar irradiance, which can be incorporated as needed.

Facilities in the LDAR program (henceforth 'sites') are provided to LDAR-Sim as geographic coordinates to establish local environmental context and topology. A single site could be used, a site list comprised of a company's assets, or all sites in a basin or jurisdiction. Custom survey frequencies can be assigned to each site for each method used in a program. These can be uniform for simple scenarios or based on facility type or risk factors such as production, age, or site-specific historical LDAR data. Survey frequencies can also be set according to accessibility. For example, if a site is not accessible by MGL, it can be designated an OGI site. The time required to survey each site can also be specified by method.

2.2. Initialization

To start the simulation, methods and site emissions are initialized. In LDAR-Sim the user can simultaneously deploy multiple methods, each with one or more independent crews. To achieve this structure, each method has a 'company' that manages daily work requirements by deploying crews. This construct also enables optimization of crew management. Each company in the program is initialized by building a deployment day matrix defining whether each site is available on each day of the simulation according to the environmental constraints set during program definition. This results in n_m matrices of dimensions n_s and t, where n_m is the number of methods in the program, n_s is the number of sites, and t is simulation timesteps in days. To register a deployment day, all environmental conditions at a site must satisfy the user-defined operational envelopes for that method.

Site emissions are initialized as fugitive and vented emissions. For each site, the number of initial leaks is randomly drawn from an empirical distribution of leak counts, and each leak is assigned an emission rate from an empirical leak-size distribution. Bias can arise if count and rate distributions are not acquired using the same detection method. For example, if counts are acquired from a study using Method 21 instruments and rates from a study using OGIs, aggregate emissions will be overestimated due to the lower sensitivity of OGIs. Ideally, both count and rate data should come from the same study and be representative of the region or assets of interest. If local data are limited or unavailable, inputs should be as representative as possible, and sensitivity should be tested.

Most LDAR programs target only fugitive emissions, but allowable vented emissions matter because methods that measure site-level emissions are often unable to distinguish between the two. Previous studies and inventories have commonly attributed >50% of site-level emissions to vents (Allen et al., 2013; Johnson et al., 2017), so it is important to consider the potentially confounding effects of these sources. To date, 'bottom-up' component-level measurements have been difficult to reconcile with contemporaneous 'top-down' site-level measurements (Brandt et al., 2014). Possible explanations for the large discrepancy include (1) inability to measure certain sources due to access restrictions (e.g. tank vents) or instrumentation limitations (e.g. compressor exhaust lacks quality measurement methods), and (2) increased chance that top-down methods capture episodic events such as blowdowns and liquid unloadings (Johnson et al., 2017; Zavala-Araiza et al., 2015). Rather than attempt to estimate site-level vented emissions (V_i) from component-level measurements, LDAR-Sim takes an empirical set of site-level measurements and bootstraps the count

(C), leak (L), and site-total (T) empirical distributions to estimate site-level vented emissions:

$$V_i = T_i - \sum_{j=1}^{C_i} L_j.$$

This procedure generates a distribution of i (typically 1000) site-level vented emission estimates. Whereas leaks are assigned to sites and remain at the same site, vents are assigned stochastically during each survey to simulate the episodic and dynamic nature of vented emissions (Vaughn et al., 2018; Johnson et al., 2019).

2.3. Updating

After initialization, LDAR-Sim evaluates the program at daily timesteps. Tracking the passage of time is of fundamental importance in LDAR-Sim and relies on empirical inputs. Each daily timestep is broken down to the minute scale. The primary time control mechanisms are: (i) the time it takes each method to survey each facility; (ii) the time it takes for agents to travel between facilities; (iii) a user-defined maximum number of work hours per day; and (iv) optional daylight requirements that calculate daylight hours for a given latitude and day of the year. Several user-defined delays can also be introduced, including reporting delays (how long it takes for survey information to become actionable), repair delays (how long it takes for a leak to be repaired once it has been reported), and minimum survey intervals.

Each day can be split into three parts: adding new leaks, finding leaks, and repairing leaks.

2.3.1. Adding new leaks

New leaks (L_n) are generated using a site-level empirical leak production rate (LPR; leaks $\cdot$ site^{-1} $\cdot$ day^{-1}) that is independent of the number of leaks already present on site:

$$L_n \sim Binomial(1,\ LPR)$$

The LPR is an empirical representation of all conditions that lead to the occurrence of leaks, including facility age, management practices, and predictive maintenance. In theory, each site can have $t + L_i$ leaks, where L_i is the initialized leak count. This is unrealistic and is made impossible in FEAST, which limits the number of leaks at a site to a finite component count. However, leak removal processes described in the following section prevent unrealistic accumulation of leaks, and we find that specifying an upper limit is unnecessary. In LDAR-Sim only one new leak can be added to a site per day. However, for a typical LPR of 0.006, there is only a 3.6×10^{-5} chance that two new leaks will arise at a site on the same day (approximately once every 65 years); we do not consider this an important concern.

2.3.2. Finding leaks

In LDAR-Sim, we distinguish between close-range methods that are able to confirm and diagnose leaks at the component scale, and screening methods that measure at larger spatial scales. Leaks must be detected at the component scale before they can be tagged for repair and enter the repair queue. This can happen in two ways: by operators during routine visits to their sites or with close-range LDAR (e.g. OGI). Methods measuring at the equipment or facility scale (e.g. MGLs, aircraft, satellites, and most drones) must flag sites for follow-up by a close-range method as they are generally unable to discern individual sources or distinguish between vented and fugitive emissions. Operator detection represents periodic inspection, maintenance, and repair outside of an LDAR program and is governed by a null repair rate (NRR). An overview is presented in S1 with further discussion in Sections 2.4.1 and 4 below.

When an LDAR program is configured, each company in the program is instructed to work each day. Companies then deploy each of their crews and leak detection occurs in accordance with the method's work practice, user-defined performance metrics, and program requirements. When a crew is deployed, workday duration is first calculated as the shorter of either (1) the number of daylight hours, or (2) the user-defined maximum workday duration for that method. Daylight hours are calculated using the `ephem` package in python, which returns civil twilight sunrise and sunset times for a given date, latitude, and longitude (Rhodes, 2011). The crew then 'chooses' and 'visits' sites until the workday ends.

Crews attempt to visit sites that have gone the longest without LDAR. A series of checks ensures that (1) the minimum interval has passed between surveys, (2) the site is not yet in compliance for the year (to avoid conducting too many surveys), and (3) environmental conditions fall within the operational envelope. If any of these conditions are not met, the crew considers another site. If no suitable sites are identified, the workday ends. If a site is identified, the crew visits it and each leak is subject to the crew's detection module. Detected leaks are tagged for repair. Once all leaks have been attempted, the time of day for the crew advances by the sum of: (1) the site-specific survey time for the crew's method, and (2) an estimate of the time required to reach the next site.

Screening method templates are similarly configured, but flag entire sites for follow-up by a close-range method. When a site is surveyed, the method's detection module is applied to the cumulative site emissions (Q_{site}), which consists of all leaks and a vented emissions estimate (V) drawn from the empirical distribution of vents:

$$Q_{site} = V + \sum_{l=1}^{n_{leaks}} Q_l$$

If a site is detected it can be flagged, but only if Q_{site} exceeds an optional user-defined follow-up threshold. The purpose of the threshold is to prevent highly sensitive screening methods from sending follow-up crews to each site they screen. A 'follow-up ratio' may also be specified, which flags sites based on relative emissions. For example, a follow-up ratio of 0.2 will flag the top 20% of surveyed sites.

2.3.3. Repairing leaks

Leaks are repaired following optional reporting and repair delays. All methods have a reporting delay separating data collection from delivery of actionable information. For screening, two reporting delays exist in the flag-tag-repair workflow as a liaising follow-up crew is required. Once tags are reported, a repair delay must pass. The repair delay can be a single value or an empirical distribution. The true distribution is unknown but may be skewed or bimodal as some leaks can be repaired immediately (e.g. tightening a valve), while others require an equipment shutdown.

2.4. Finalization

When all simulation days are complete, LDAR-Sim outputs a series of data files, maps, and figures. Data are output by site, leak, and day to enable detailed and transparent evaluation of results (Table S1). The proportion of sites available is a global measure of suitability reflecting the daily proportion of all sites that fall within each method's environmental constraints. Although cost is not presently treated thoroughly in LDAR-Sim, users are encouraged to build their own costing modules with real data. Our placeholder cost variable is a crude estimate of daily company costs and is charged for each crew deployed.

When multiple methods are deployed, it is important to track different kinds of detection redundancy that may arise. Redundant tags are tagged leaks that are detected a second time by an operator or close-range method. These tags are not counted because the leaks have already been tagged and are awaiting repair. Three types of redundant flags exist and are described in Fig. S1. Tracking redundancy is important to ensure that multiple companies or crews are not taking credit for emissions reductions resulting from the repair of the same leak.

2.4.1. Measuring equivalence

In principle, two LDAR programs could be compared using various metrics. Examples include average emission rates, active and repaired leak counts, missed leaks, average duration between leak origin and repair, and so on. Here, our focus is on emissions reduction equivalence, the metric commonly prescribed by regulators. Various approaches can be used to estimate emissions reductions. In FEAST, emissions under an LDAR program are subtracted from a null scenario – 'business as usual' in the absence of LDAR (Ravikumar and Brandt, 2017). The null scenario balances LPR and NRR in the absence of LDAR to achieve steady-state emissions centered on the initialized state of the leak field.

We find the notion of a null scenario problematic for several reasons. First, the 'pre-LDAR' baseline is elusive, and uncertainty in its estimate propagates to mitigation. Today, many companies have some kind of leak management program already in place. In Canada, a shift towards more formalized LDAR occurred in 2007 when best management practices were published by an industry group (CAPP, 2007). Second, fugitive emissions are unlikely to be in equilibrium. Management practices, new technology, and pressure on producers to reduce emissions may result in a downward trend in fugitive emissions. While both possibilities are speculative, the burden of proof should be on equilibrium if this assumption is required. Third, the null baseline is maintained by LPR and NRR, two poorly understood parameters. Both LPR and NRR have significant challenges:

- LPR has never been estimated independently of NRR; in deriving LPR, it is assumed that NRR and LDAR are inconsequential. This is inconsistent with assumptions made during simulation, where NRR is assigned a value approximately equal to LPR. If estimating an LPR that already encodes null repair, NRR should not need to be applied during simulation.
- LPR is most easily estimated from data acquired during LDAR surveys, compromising the 'pre-LDAR' steady state assumption.
- While attempts have been made to estimate LPR, NRR has neither been measured nor empirically inferred.
- In FEAST, NRR is a static value defined using LPR. The same NRR is used for all leaks, regardless of emissions rate. However, large leaks may be more easily detected by human senses, and if discovered, are of greater safety and financial concern. If removal rate is a function of leak size, production of larger leaks must increase to maintain the relative proportion of large leaks.
- NRR likely depends on company culture and individual operator practice. In FEAST, a higher LPR results in a higher NRR. However, if company culture matters, LPR and NRR may be inversely related.

Given these challenges, it is conceivable that our best estimates for LPR and NRR are wrong, perhaps by a large margin. We therefore avoid the concept of a null scenario and compare programs directly based on emissions instead of mitigation. This can be formalized: Let equivalence between two programs (P_1 and P_2) be defined as equal mitigation ($M_1 = M_2$). Assume that $M = Null_E - P_E$, or emissions from the null scenario minus program emissions. If the null scenario is the same for both programs, equivalence can be defined as $Null_E - P_{E1} = Null_E - P_{E2}$, where $Null_E$ cancels out.

3. Case studies

To demonstrate LDAR-Sim, we present two case studies of LDAR programs deployed in Alberta, Canada. In the first case study, two alt-LDAR programs are compared to a hypothetical reg-LDAR program. The first alternative program (P_1) uses quarterly MGL screening combined with follow-up OGI at 80% of facilities emitting more than 100 g h^{-1} (Table 1). The second alternative program (P_1) requires aircraft screening 6 times per year with follow-up OGI at 50% of the facilities emitting more than 5000 g h^{-1} (Table 1). The reference reg-LDAR program (P_{Ref}) requires triannual OGI at all facilities. In the second case study, P_{Ref} is compared to three additional OGI-based programs (P_W, P_L, and $P_{W,L}$) to explore how reg-LDAR program performance can vary with environmental and labour constraints. The three new OGI programs differ from P_{Ref} in either (i) temperature, wind, and precipitation deployment thresholds (P_W and $P_{W,L}$), or (ii) the number of deployment crews available to do work (P_L and $P_{W,L}$). P_W, P_L, and $P_{W,L}$ are otherwise identical to P_{Ref}. All case study programs presented in this paper are hypothetical and intended only for demonstration.

All case study programs are implemented over a 7-year period (2011–2018; using historical weather data) alongside the OGI-based regulatory standard required by Environment and Climate Change Canada (ECCC) (Government of Canada, 2018). Briefly, ECCC requires triannual surveys on most upstream production facilities, with the exception of single wellheads with no associated equipment (i.e. connected directly to a gathering pipeline). Most leaks must be repaired within 30 days, unless a major shutdown is required. We demonstrate these programs for a producer with 1169 Alberta facilities requiring LDAR. To maintain anonymity of the producer, random noise $\mathcal{N}(0, 0.1)$ was added to the x and y coordinates of each site and no identifying information is reported. This has negligible effect on the conditions polled from weather data. The sites are broadly representative of Alberta's diverse geography and producing regions, spanning approximately 9.6° of longitude and 8.7° of latitude. For each program we conduct 25 simulations over 2557 daily timesteps and assume a standard repair delay of 14 days for all leaks. Input parameters are summarized in Table 1 and explained in S2.

Case study results are presented in Fig. 2 and Table S2. In general, emissions in Fig. 2 increase according to the LPR and decrease when LDAR is conducted. Each program configuration leads to distinct emissions trends (Fig. 2). Only P_1 achieves equivalence to the reference case. Although less sensitive (the OGI median detection limit is ~10 g h^{-1}), P_1 measures cumulative site-level emissions, including vents, and sends follow-up crews to 80% of detected sites. In this configuration, P_2 is not equivalent to P_{ref} as it is only visiting sites emitting above 5000 g h^{-1}. Small leaks are still eventually resolved, as follow-up crews conduct comprehensive surveys when tasked to a high-emitting site. Despite large differences in emissions (and therefore mitigation), each of the programs results in roughly the same number of tagged leaks (Table S2). This suggests that screening methods may be less efficient than close-range methods, because (1) the presence of vented emissions can influence follow-up decisions, and/or (2) higher detection limits require more leaks to build up before a follow-up is triggered.

Periodicity in emissions is driven by survey frequency, labour availability, and deployment constraints. The slope of emission reductions relates to the size of the workforce relative to the number of facilities to be surveyed. In FEAST, this slope is vertical, creating a 'saw tooth' emissions profile not observed in LDAR-Sim because agents typically require several weeks or months to

Table 1
Parameters used in the two case studies. Case study 1 compares alt-LDAR programs P_1 and P_2 to reg-LDAR program P_{Ref}. Case study 2 compares a range of OGI-based programs (P_{Ref}, P_W, P_L, and $P_{W,L}$). These programs are hypothetical and based on fabricated detection methods and survey requirements for the sake of demonstration. See S2 for detailed explanation of case study parameterizations.

	P_{Ref}	P_W	P_L	$P_{W,L}$	P_1[a]	P_2[a]	[b]
Method	OGI	OGI	OGI	OGI	MGL	Aircraft	OGI
Number of crews	3	3	1	1	1	1	2
Minimum temp (°C)	−30	0	−30	0	−30	−30	−30
Maximum wind speed ($m \cdot s^{-1}$)	20	5	20	5	20	20	20
Maximum precipitation (m)[c]	0.01	0.001	0.01	0.001	0.01	0.01	0.01
Minimum repeat interval (days)	120	120	120	120	50	50	–
Maximum workday (hours)	10	10	10	10	10	10	10
Reporting delay (days)	2	2	2	2	2	2	2
Detection limit ($g \cdot hour^{-1}$)	A	a	a	a	100	5000	a
Time offsite ($minutes \cdot site^{-1}$)	B	b	b	b	b	10	b
Required surveys per site	3	3	3	3	4	6	–
Time per site (minutes)	120	120	120	120	30	5	120
Follow-up threshold ($g \cdot hour^{-1}$)	–	–	–	–	100	5000	–
Follow-up ratio	–	–	–	–	0.8	0.5	–

[a] Configured using Ravikumar et al. (2018). See text for details.
[b] Drawn from an empirical distribution. See text for details.
[a] Screening programs include two OGI follow-up crews.
[b] This column gives relevant follow-up parameters for screening methods.
[c] Accumulated between 12:00 and 18:00.

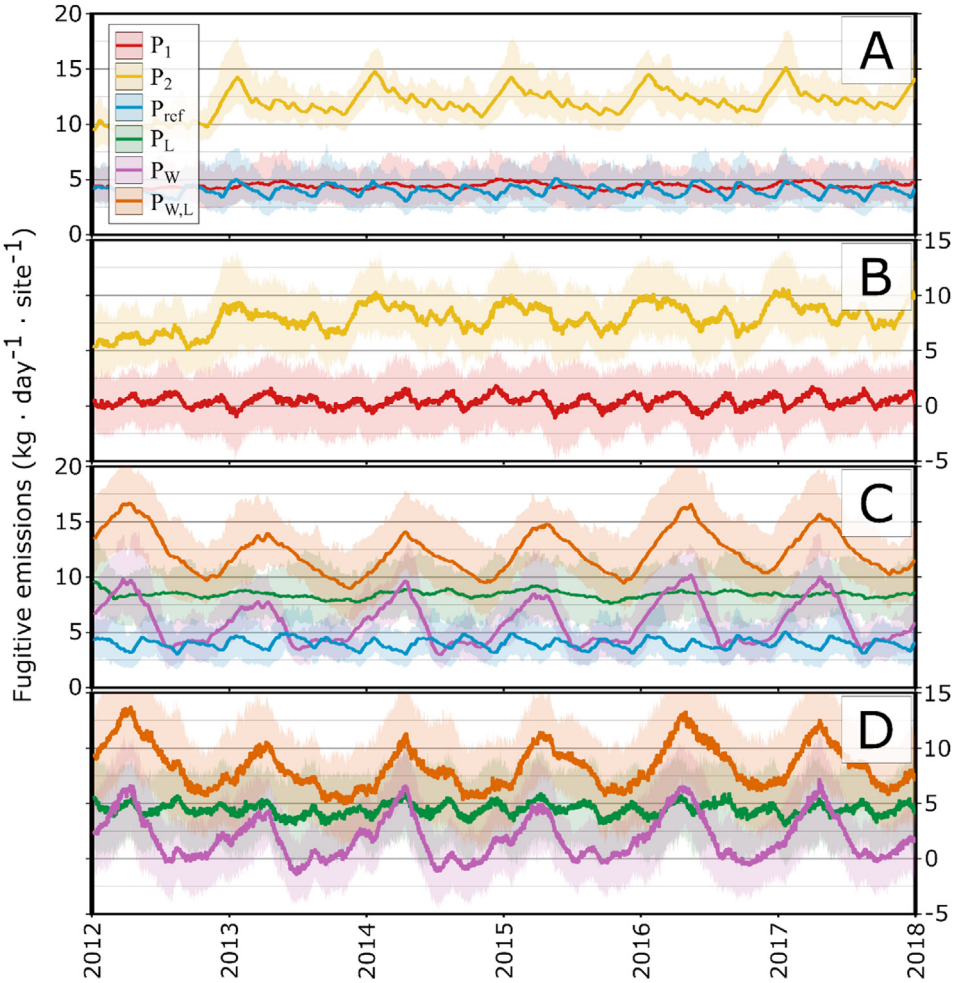


Fig. 2. Fugitive emissions observed under each program. Lines show daily emissions averaged across 25 simulations (±2σ). A and C show average site-level emissions, while B and D show candidate programs differenced against the regulatory program ($P_X - P_{ref}$). In B and D, equivalence occurs at y = 0. See Table S2 for time-averaged results and confidence intervals.

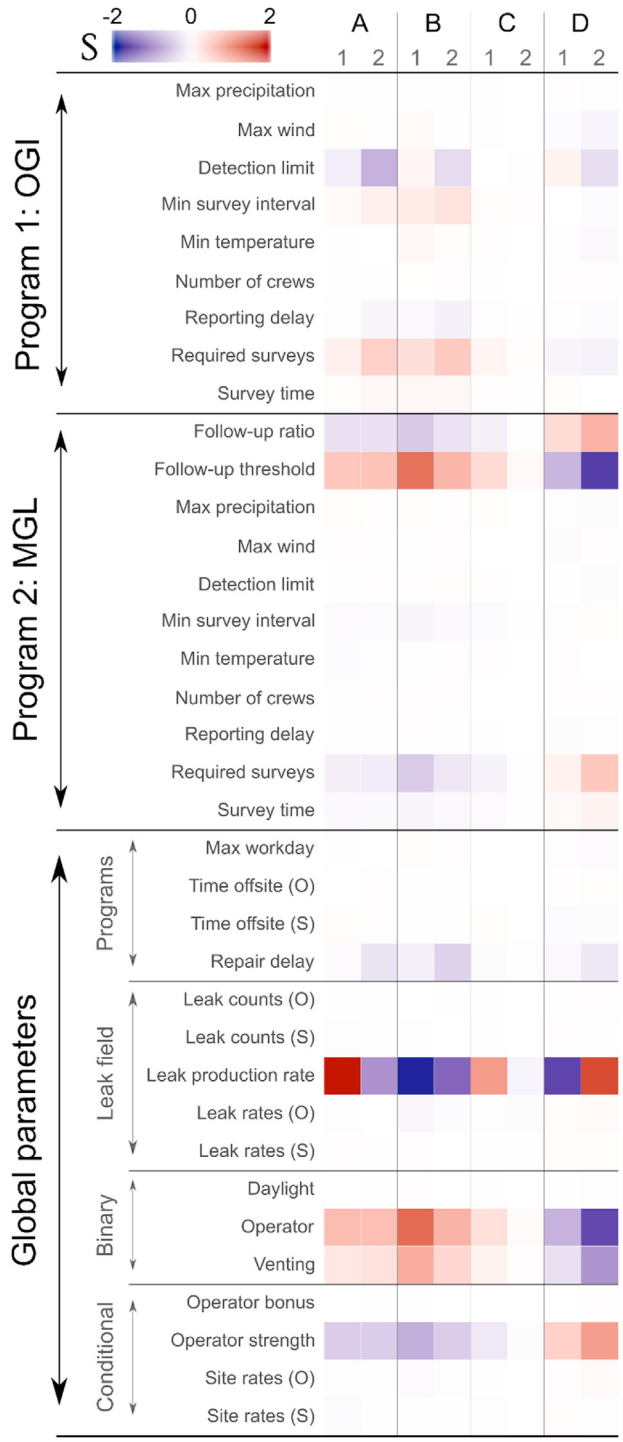


Fig. 3. Sensitivity analysis results. In general, equivalence determinations are more likely to depend on inputs as S departs from zero (darker colours). Values of S near zero (lighter colours) are ideal as they suggest that equivalence metrics are similar across inputs. Values near zero don't necessarily indicate that each individual program is unaffected by the input, but that the relative difference between two programs will be similar for any input value. Equivalence metric outputs are: (A) median number of active leaks, (B) median number of days leaks remain active before discovered, (C) mean daily site emissions, (D) cumulative number of repaired leaks. For each output, column 1 calculates S using absolute differences between programs (MGL-OGI), while column 2 uses a ratio (MGL/OGI). The S index approximates the number of standard deviations separating equivalence metrics from the highest and lowest input parameters (see S3 for full derivation of S). For example, high values (e.g., S > 1 and red) suggest that when an input parameter is high, the MGL program produces higher outputs relative to the OGI program than at low input values. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

complete each round of surveys. The absence of periodicity in P_L and P_1 suggests that agents are working continuously. The tri-annual fluctuations in P_{ref} suggest that crews survey all required sites before the next round of surveys is required. A balance must be found between having idle crews and a risk of falling out of compliance. Strong periodicity can be indicative of an oversupply of labour or of operational envelopes restricting surveys to seasonal clusters. The weather-restricted programs (P_W and $P_{W,L}$) exhibit strong seasonal (wintertime) periodicity resulting in ~50% higher fugitive emissions (Fig. 2C, Table S2). To better understand how weather affects deployment days, LDAR-Sim generates maps to explore the spatial distribution of suitable conditions for a given set of environmental constraints (Fig. S2).

The variable performances of the OGI-based programs in Fig. 2C illustrate the importance of defining the reference program for determining equivalence. Most jurisdictions require demonstration that alt-LDAR is equivalent to reg-LDAR but are not explicit about the quantity of emissions permitted or mitigated. When P_2 is compared to P_{ref} it does not achieve equivalence. However, P_2 is closer to $P_{W,L}$ and could be deemed equivalent. To achieve defensible definitions for emissions reductions, regulators should be explicit in delineating permissible contexts or expected mitigation for equivalence demonstration.

4. Model confidence

Building confidence in LDAR models is challenging because (1) many of the parameters used to characterize leak fields, detection methods, and repair remain poorly constrained, (2) empirical inputs, and therefore results, vary in time and space, and (3) validation capacity is limited by the high cost of LDAR programs, the long periods over which they can be implemented, and the challenge of accurate emissions measurement. Although the value of these tools is heuristic, model confidence can be established by comparing similar models, conducting SA, and comparing against real LDAR programs.

The only model that can be used for comparison is FEAST, which has similar leak field properties but differs in most other aspects. Results from FEAST simulations of OGI-based LDAR programs are presented in two studies (Kemp et al., 2016; Ravikumar and Brandt, 2017). Both use the same LPR as our case study, but all three studies use different empirical leak distributions. In the first, fugitive emissions fluctuate between ~1 and 10 $kg \cdot day^{-1} \cdot site^{-1}$ for a survey interval of 100 days (Kemp et al., 2016). In the second, quarterly and annual surveys result in mean emissions rates of ~13 and 26 $kg \cdot day^{-1} \cdot site^{-1}$, respectively (Ravikumar and Brandt, 2017). Results from LDAR-Sim's OGI demonstration fall within these two studies (~3–13 $kg \cdot day^{-1} \cdot site^{-1}$; Fig. 2), although neither model has been validated against field measurements.

To explore how different parameterizations could impact equivalence determinations, we compare an OGI-based program to a generic MGL screening program in an extensive SA (see S3 for methods). We compare the two LDAR programs using both differences and ratios of their outputs to explore which method provides more robust equivalence metrics. Results show that leak production rate is the most sensitive input parameter (Fig. 3), similar to past results (Kemp et al., 2016). As LPR increases, the absolute difference in emissions between programs increases (Fig. 4a), with the OGI program outperforming the screening program. However, relative emissions between programs appear independent of LPR ($S = -0.3$ for ratio and 1.7 for absolute difference; Fig. 3), suggesting

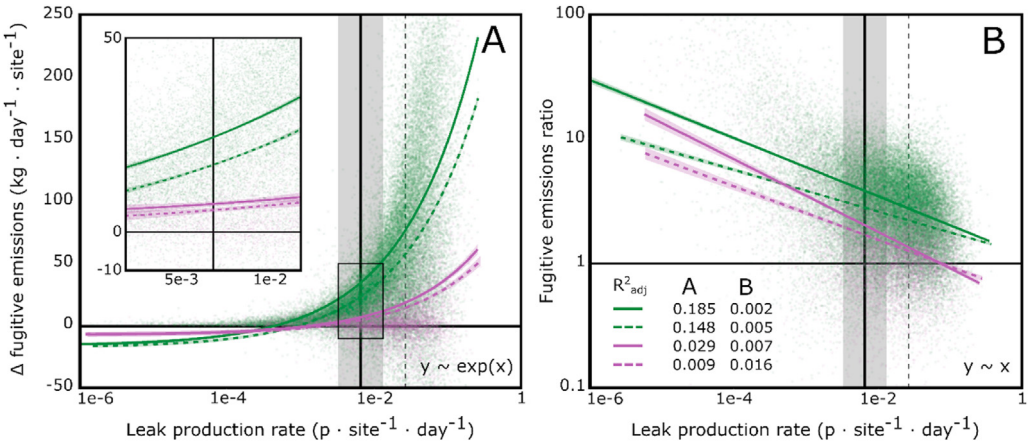


Fig. 4. Simulated comparison (n = 50 000) of two LDAR programs using differences (A; MGL - OGI) and ratios (B; MGL/OGI) of fugitive emissions as a function of LPR. Each dot compares a close-range and a screening program that share global parameters. Purple and green lines show operator presence and absence, respectively (i.e. NRR turned on or off). Dashed lines show presence and solid lines show absence of vented emissions. The dashed vertical line is the mean LPR from the SA sampling distribution (0.026), established using Alberta data. The solid vertical line is the LPR value used in FEAST and the shaded region is the LPR SA range evaluated in Kemp et al. (Kemp et al., 2016) Adjusted R^2 values near zero are preferred, as these indicate a lower dependence of emissions equivalence on LPR. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

that detection capacity of different programs may scale with total emissions, despite falling behind in absolute terms. Given that LPR remains an elusive parameter, we recommend comparing programs on a relative rather than absolute basis. If absolute differences in emissions are used, establishing equivalence may be easier for low-LPR companies or regions. Future research of repeat field surveys should feature robust record keeping of repairs and leak history (new, re-occurred, and not repaired) to improve LPR estimates.

Follow-up work practices and the presence of venting impact the performance of screening programs (Fig. 3). Screening methods that measure site-level emissions are unable to distinguish between vented and fugitive sources. As vented emissions increase, follow-up decisions may preferentially target high-venting sites at the expense of fugitive emissions. The impact of venting on relative performance is higher when LPR is low (Fig. 4b), likely because fugitive emissions are lower and more likely to be overwhelmed by vented emissions, which are independent of LPR. The SA suggests that increasing follow-up ratios and decreasing thresholds can improve screening performance. For alt-LDAR programs to be appealing to producers, trade-offs between follow-up rules and deployment costs should be considered.

Results are sensitive to assumptions surrounding operator activity (Fig. 3). As more leaks are repaired by operators, fewer remain for LDAR programs. Higher operator activity therefore results in smaller differences between programs, making demonstration of equivalence easier. The operator effect tends to increase with LPR, but mostly when comparing absolute differences in outputs. Given the evidence available, we advocate for disabling NRR in most situations, because (1) most empirical LPR estimates implicitly encode NRR, making additional NRR activity redundant; (2) no empirical NRR data exist and AVO detection metrics have not been established; (3) the assumption that system-wide fugitive emissions are in equilibrium lacks evidence; and (4) incorporating NRR obscures program comparison, increasing the threat of erroneously declaring two programs equivalent. Although LPR and NRR are nearly impossible to measure separately, they are relatively easy to measure together. Rather than attempt to estimate both LPR and NRR independently, we recommend modelers (1) reject the requirement of a steady-state and a null scenario and (2) work only with the empirical LPR that already accounts for NRR.

Several input variables are of seemingly low influence (Fig. 3). For example, LDAR-Sim outputs do not appear to be sensitive to the addition or removal of super-emitters in any of the empirical input distributions. Others include operator bonus (p_{max}), detection limits for MGLs, and leak-size distributions. However, these inputs may still impact the performance of individual programs (e.g., Fig. S5). For example, increasing the number of super-emitters can increase program emissions, but if the increase is approximately proportional for both programs, their relative performance remains similar and S will be near zero. Additional variables and functionality not considered in our case study or SA may be important, depending on program configuration. Targeted SAs are therefore necessary each time LDAR-Sim is used, as input variables may become sensitive when tested against a narrower range of other inputs.

5. Conclusion

At present, collective understanding of how to configure LDAR programs is nascent. Many of the inputs required to define LDAR programs are poorly constrained. Detection modules presented here are based on hypothetical performances and are not tied to specific work practices. To be used effectively, custom modules should be developed for each candidate method. These modules should reflect each method's unique set of performance metrics developed through controlled testing. Field data will improve understanding of these variables and identify others that may warrant inclusion. By means of open-source development, LDAR-Sim can evolve alongside this knowledge base, and may grow to be used in different ways by regulators, producers, consultants, solution providers, technology developers, investors, and non-profits.

Regulators can play an important role in this transition and enabling the success of tools like LDAR-Sim. Not all jurisdictions have introduced policies to mandate monitoring and reduction of fugitive methane emissions. Of those that require LDAR, few permit alt-LDAR. However, interest is growing rapidly and more alt-LDAR policies are expected soon. As new policies emerge, regulators should work together to establish clear requirements and standardized protocols for demonstrating equivalence, especially where modeling is involved. Regulators should also work to clearly and quantitatively define equivalence based on emissions reduction targets and expected performance of approved methods. If authorizing the use of LDAR-Sim, regulators should also be

intimately familiar with the ways it might be used by applicants. Ultimately, LDAR-Sim is one tool that can be used as part of a broader effort to identify emerging LDAR methods and should be supported with empirical data and expert advice.

LDAR-Sim is a modular, open-source modeling framework that can be further developed as new functionality is required or new measurement methods are developed. Future work will identify alt-LDAR programs that improve upon reg-LDAR, track fugitive and design emission with high spatial and temporal resolution, develop methods to optimize cost to mitigation ratios, validate model predictions through field trials, and improve integration of empirical inputs. More flexibility in spatial resolution should be built into LDAR-Sim, as some methods monitor at the equipment scale, providing more detailed information than facility-scale screening. Other possible improvements include modeling predictive maintenance efforts, quantification accuracy of screening methods, additional environmental variables to define operational envelopes of new methods, and a more sophisticated reproduction of repairs.

Author contributions

TAF, MG, TEB, and CHH contributed to project conception and design. TAF, MG, and TEB designed and wrote the model code. TAF conducted the case study, including parameterization, simulation, and analysis. TAF designed and conducted the sensitivity analysis and drafted the manuscript. All authors contributed to writing, review and approval of the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations

alt-LDAR	Alternative LDAR
AVO	Audial, visual, and olfactory
ECCC	Environment and Climate Change Canada
FEAST	Fugitive emissions abatement simulation testbed
LDAR	Leak detection and repair
LPR	Leak production rate
MGL	Mobile ground laboratory
NRR	Null repair rate
OGI	Optical gas imaging
reg-LDAR	Regulatory LDAR
SA	Sensitivity analysis
US	United States

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2020.125237.

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